

Applied Thermodynamic II

ME2402

Chapter 01

Heat Transfer



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Content

- Introduction to Heat Transfer
- Fourier's Law of Conduction
- Newton's Law of Cooling
- Steady State Heat Flow through Composite Wall
- Cylinder & Sphere

1.1 Introduction to Heat Transfer

What is heat transfer

- Heat is a form of **energy** which transfer from one body to another body at **low temperature**.
- This is called as **Natural Heat Transfer**



A: Ceramic cup



B: Aluminum cup



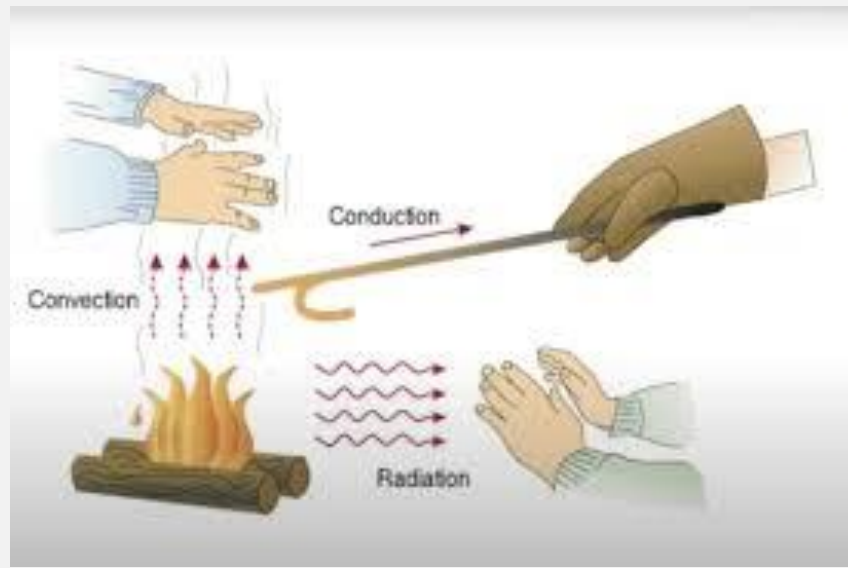
C: Coconut shell



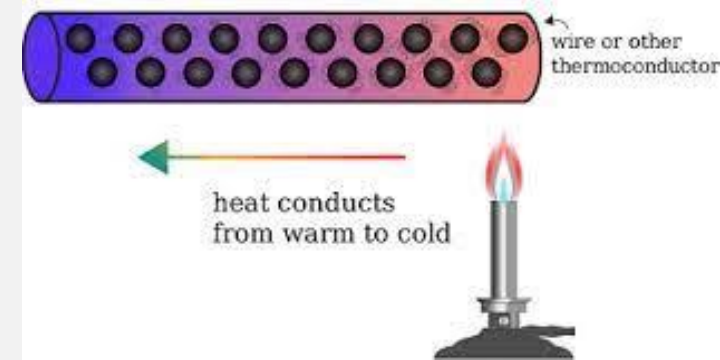
1.1 Introduction to Heat Transfer

Modes of heat transfer

- Conduction : heat transfer between two objects by direct contact
- Convection : heat transfer from movement of fluid
- Radiation : heat transfer using heat waves (electromagnetic)



1.2 Conduction Heat transfer (1D)



- Heat transfer from one part of substance to another part or to another substance in physical contact, where **displacement of molecule is negligible**.

Fourier's law of conduction

- This law govern the conduction heat transfer

$$\dot{Q} \propto A \frac{dt}{dx}$$

Where ;

$\dot{Q}$ is the rate of heat flow in x direction

A is the surface area (should be **Perpendicular**

$\frac{dt}{dx}$ is change of temperature respect to length (temperature gradient)

1.2 Conduction Heat transfer

Fourier's law of conduction

Where ;

$\dot{Q}$ is the rate of heat flow in x direction

A is the surface area (should be parallel to x-dir)

$\frac{dt}{dx}$ is change of temperature respect to length or temperature gradient

k is the thermal conductivity (W/mK)

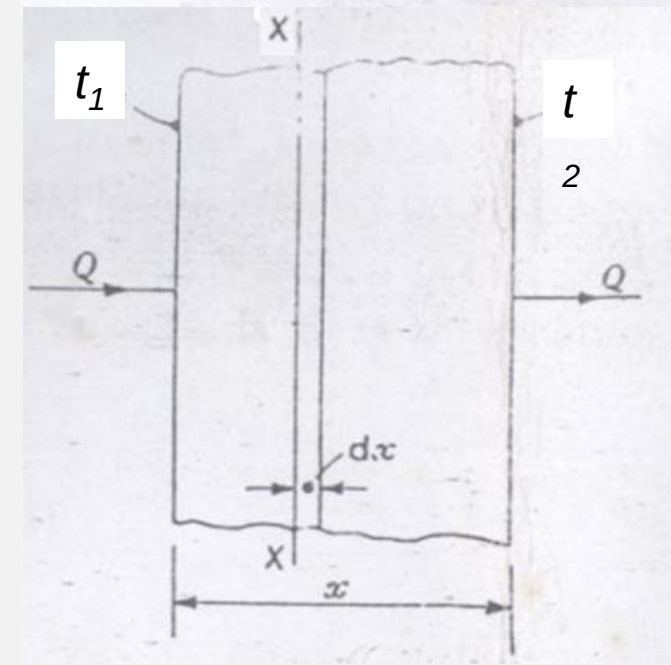
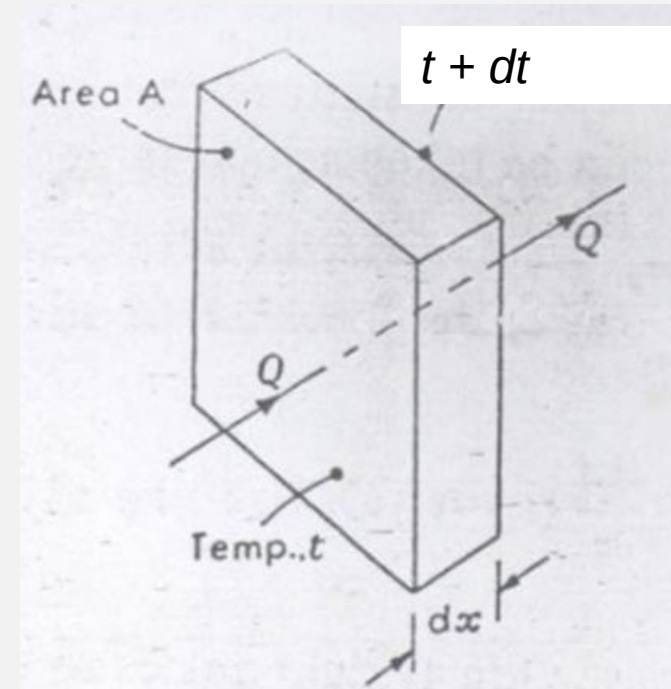
$$\dot{Q} \propto A \frac{dt}{dx}$$

$$\dot{Q} = -kA \frac{dt}{dx}$$

Do some integrations.....

Finally;

$$\dot{Q} = kA \frac{(t_1 - t_2)}{x}$$



1.2 Conduction Heat transfer

Fourier's law of conduction

$$\dot{Q} \propto A \frac{dt}{dx}$$
$$\dot{Q} = -kA \frac{dt}{dx}$$

Where ;
 $\dot{Q}$ is the rate of heat flow in x direction
 A is the surface area (should be parallel to x-dir)
 $\frac{dt}{dx}$ is change of temperature respect to length or temperature gradient
 k is the thermal conductivity (W/mK)

Material	Thermal Conductivity (W/mK)
Diamond = 2200	
Pure Copper	378
Pure Aluminum	206
Steel	43
Brick,common	0.17 – 0.34
Concrete	0.87 – 1.38

Material	Thermal Conductivity (W/mK)
Glass	0.87
Glass fiber insulation	0.05
Ice	2.2
Plastic	0.17
Wood	0.17 – 0.34

CorkBoard = 0.043

1.2 Conduction Heat transfer

Fourier's law of conduction

Example 01

The inner surface of a plane brick wall is at 40°C and the outer surface is at 20°C . Calculate the rate of heat transfer per unit area of wall surface; the wall is 250 mm thick and the thermal conductivity of the brick is 0.52 W/m K .

Answer 41.6 W/m^2

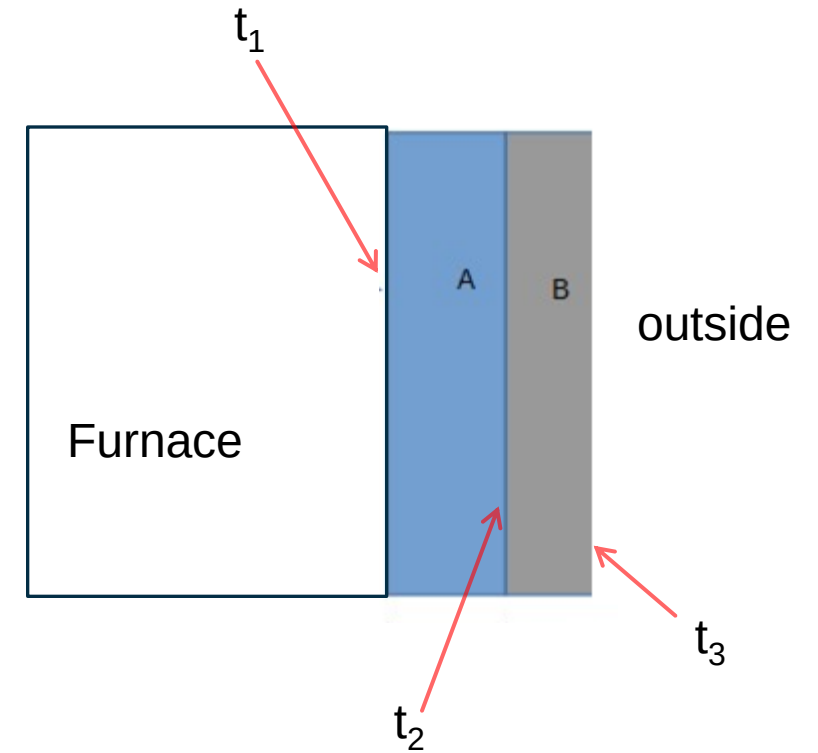
1.2 Conduction Heat transfer

Fourier's law of conduction

Example 02

Furnace wall consists of a refractory brick layer (A) and a plaster layer (B) as shown in the image. The temperature of inside furnace wall is 1100°C which is as t_1 in the image. The temperature at inner joint is 995°C and given as t_2 in the image. The thermal conductivity of refractory brick is 1.6 W/mK and 0.14 W/mK . Thickness of the refractory brick and plaster layer are 125mm and 12mm respectively.

- Calculate the rate of heat loss per unit area of wall surface
- Calculate the temperature of the outer surface(t_3)



Answer 1.344 kW , 879.8°C

Are you satisfied with the above answers ?

Suggestions to improve ?

1.2 Conduction Heat transfer

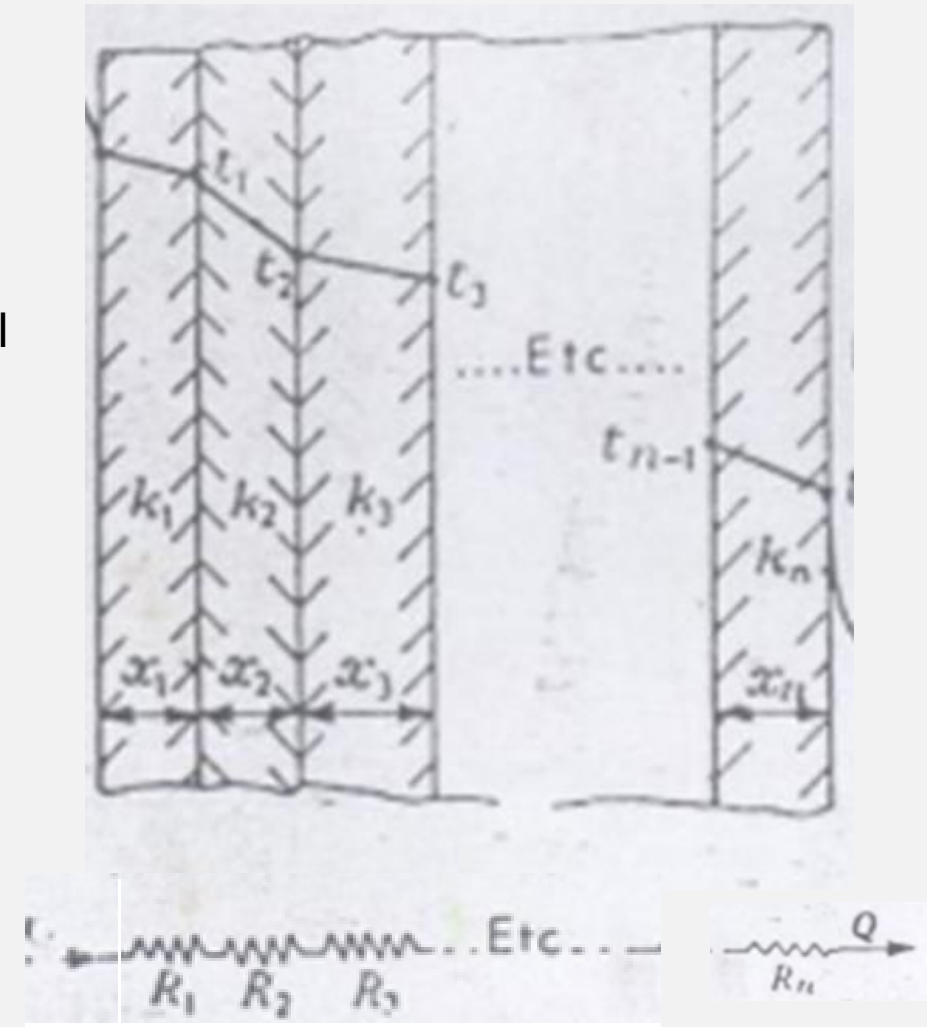
Electrical analogy for composite wall etc..

- Composite walls or typical insulation materials are made with several layers
- It is difficult to do the calculations in the previous method
- So we use electrical analogy for such situations

Heat Flow (Q) > Analogous > Electric Current (I)

Temperature Difference ($t_1 - t_2$) > analogous > Potential Difference (V)

Thermal Resistance > analogous > Electrical Resistance (R)



1.2 Conduction Heat transfer

Electrical analogy for composite wall etc..

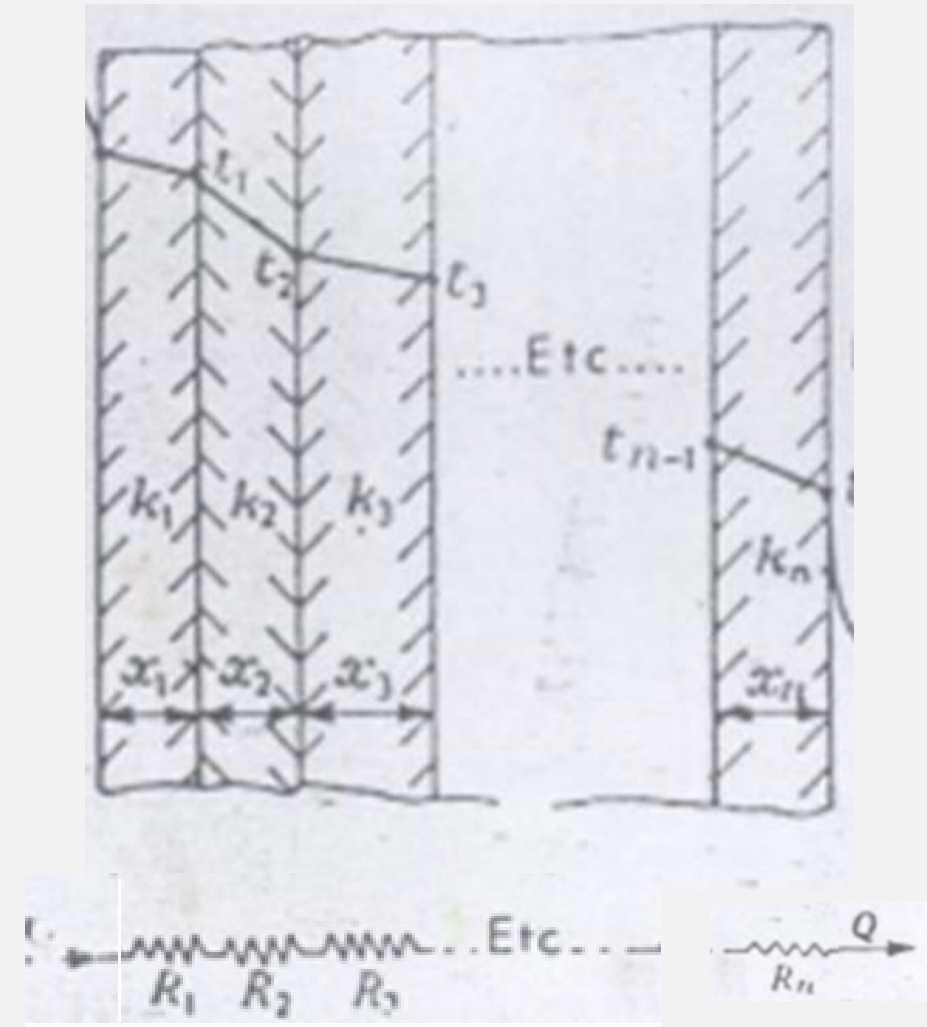
Heat Flow (Q) > Analogous > Electric Current (I)

Temperature Difference ($t_1 - t_2$) > analogous > Potential Difference (V)

Thermal Resistance > analogous > Electrical Resistance (R)

So, the Thermal Resistance (R);

$$R = \frac{x}{kA}$$



1.2 Conduction Heat transfer

Electrical analogy for composite wall etc..

Example 03

Home work

